MA 5755: Data Analysis & Visualization in R/Python/SQL

Week 10: Classification and Support Vector Machines

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From Regression to Classification

We now move from regression to classification.
Given training data:

$$\{(x_i, y_i)\}_{i=1}^{n}, \quad y_i \in \{1, \ldots, K\}$$

Goal: learn a decision rule:

$$G(x) : \mathbb{R}^p \to \{1, \ldots, K\}$$

Key idea:

- Regression estimates $\mathbb{E}[Y|X]$
- Classification estimates $P(Y = k|X)$

Bayes classifier:

$$G^*(x) = \arg\max_k P(Y = k|X = x)$$

Linear Regression for Classification

Encode classes using indicator vectors:

$$Y \in \mathbb{R}^K, \quad Y_k = \mathbf{1}(Y = k)$$

Fit:

$$\hat{f}(x) = Wx$$

Prediction:

$$\hat{G}(x) = \arg\max_k \hat{f}_k(x)$$

Problem:

- Outputs not constrained to $[0, 1]$
- Poor probabilistic interpretation

So, it seems that we need better models.

An illustration

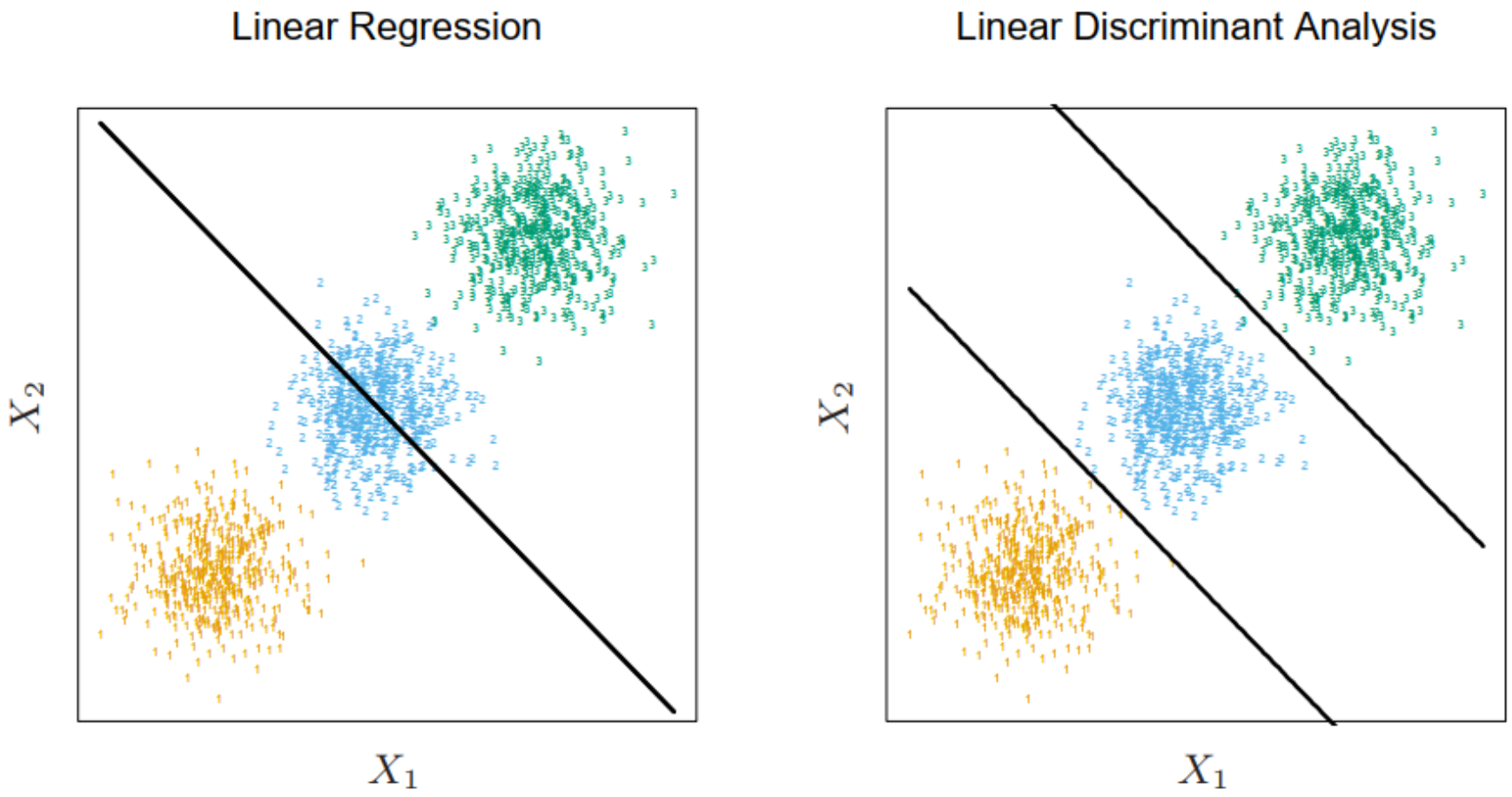


FIGURE 4.2. *The data come from three classes in $\mathbb{R}^2$ and are easily separated by linear decision boundaries. The right plot shows the boundaries found by linear discriminant analysis. The left plot shows the boundaries found by linear regression of the indicator response variables. The middle class is completely masked*

More details: An illustration

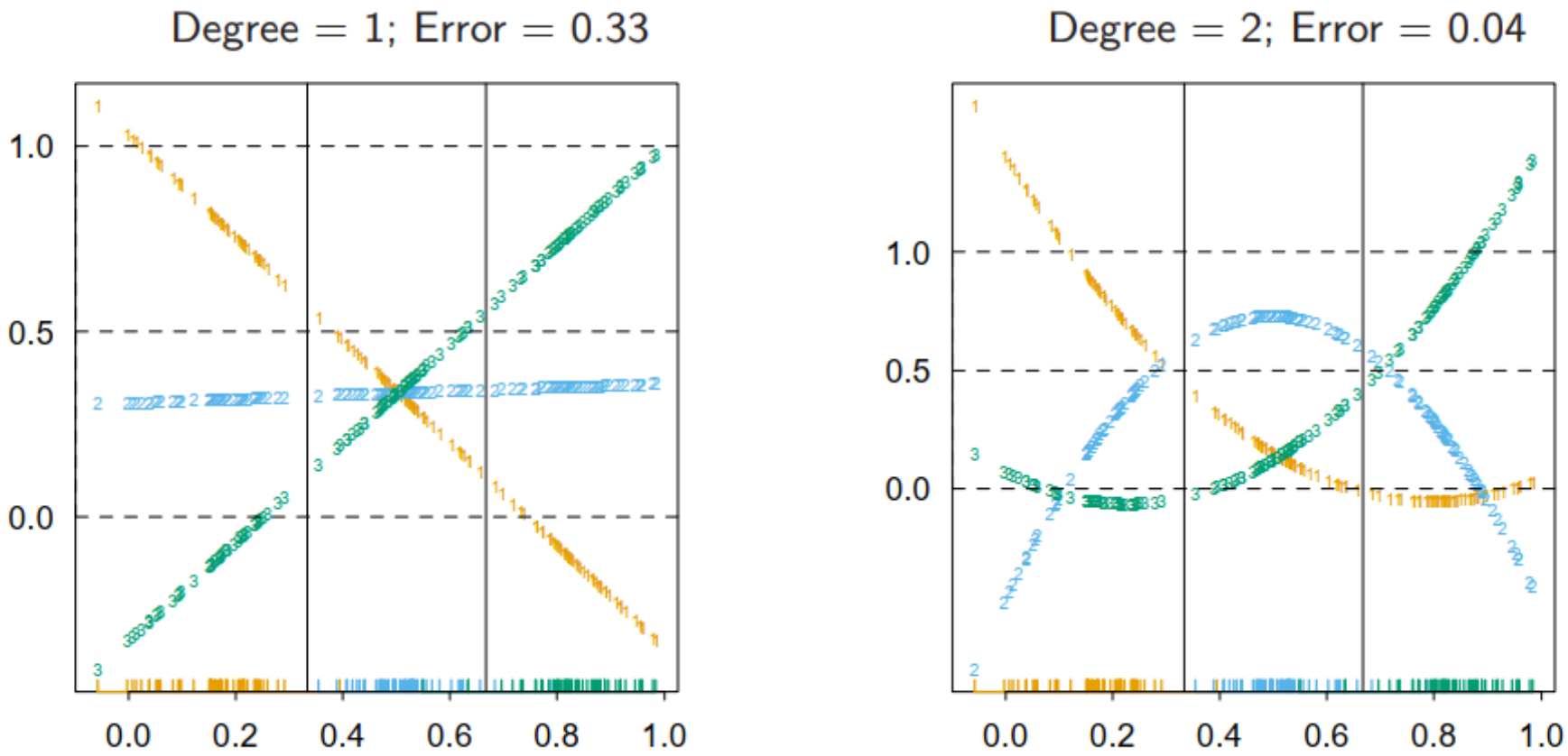


FIGURE 4.3. *The effects of masking on linear regression in* $\mathbb{R}$ *for a three-class problem. The* rug plot *at the base indicates the positions and class membership of each observation. The three curves in each panel are the fitted regressions to the three-class indicator variables; for example, for the blue class,* y_{blue} *is* 1 *for the blue observations, and* 0 *for the green and orange. The fits are linear and quadratic*

Generative vs Discriminative Models

Two general types of approaches:

1. Generative: Basically asks "How was this data produced?" Models the data-generating process. Given $P(X|Y), P(Y)$, find

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

2. Discriminative: Basically asks "Given this data, what is the label?" Draws the boundary needed to classify.

$$\text{Models} \quad P(Y|X) \quad \text{directly}$$

Examples:

- Linear Discriminant Analysis: generative
- Logistic regression: discriminative

Linear Discriminant Analysis (LDA)

LDA assumes that each class generates data from a Gaussian distribution with a common covariance, so classification is based on comparing how likely a point is under each class distribution.

$$X|Y = k \sim \mathcal{N}(\mu_k, \Sigma)$$

Using Bayes rule:

$$\delta_k(x) = \log(P(Y = k|X = x)) = x^T\Sigma^{-1}\mu_k - \frac{1}{2}\mu_k^T\Sigma^{-1}\mu_k + \log \pi_k$$

Here π_k is the prior probability of class k. Also, parameters need to be estimated: π_k, μ_k, Σ.
The model reduces classification to evaluating linear scoring functions.
Decision rule:

$$G(x) = \arg\max_k \delta_k(x)$$

Geometry of LDA

Decision boundary between classes k and l: $\delta_k(x) = \delta_l(x)$ leads to

$$w^T x + b = 0 \quad \text{where} \quad w = \Sigma^{-1}(\mu_k - \mu_l)$$

Intuition and implications:

- Decision boundary is linear because classification depends only on differences between class means under a shared covariance structure.
- w shows that LDA projects data onto a direction that maximizes separation relative to within-class variability.
- The transformation Σ^{-1} whitens the space, so distances are measured in units of variance rather than raw Euclidean distance.
- The classifier effectively measures distance to class means after scaling by Σ^{-1}, so directions with low variance are given higher importance.

Intuition and implications (contd.)

- Classification is projecting data onto a single direction and thresholding, which reduces a high-dimensional problem to 1-D.
- Thus, LDA performs well when classes are well separated along a dominant direction, but may fail when separation is nonlinear.
- Because the model is fully probabilistic, it produces class posterior probabilities and not just hard decisions.
- Strong distributional assumptions lead to low variance and stable estimates, especially in small-sample settings, but can introduce bias if the Gaussian assumption is violated.

An example

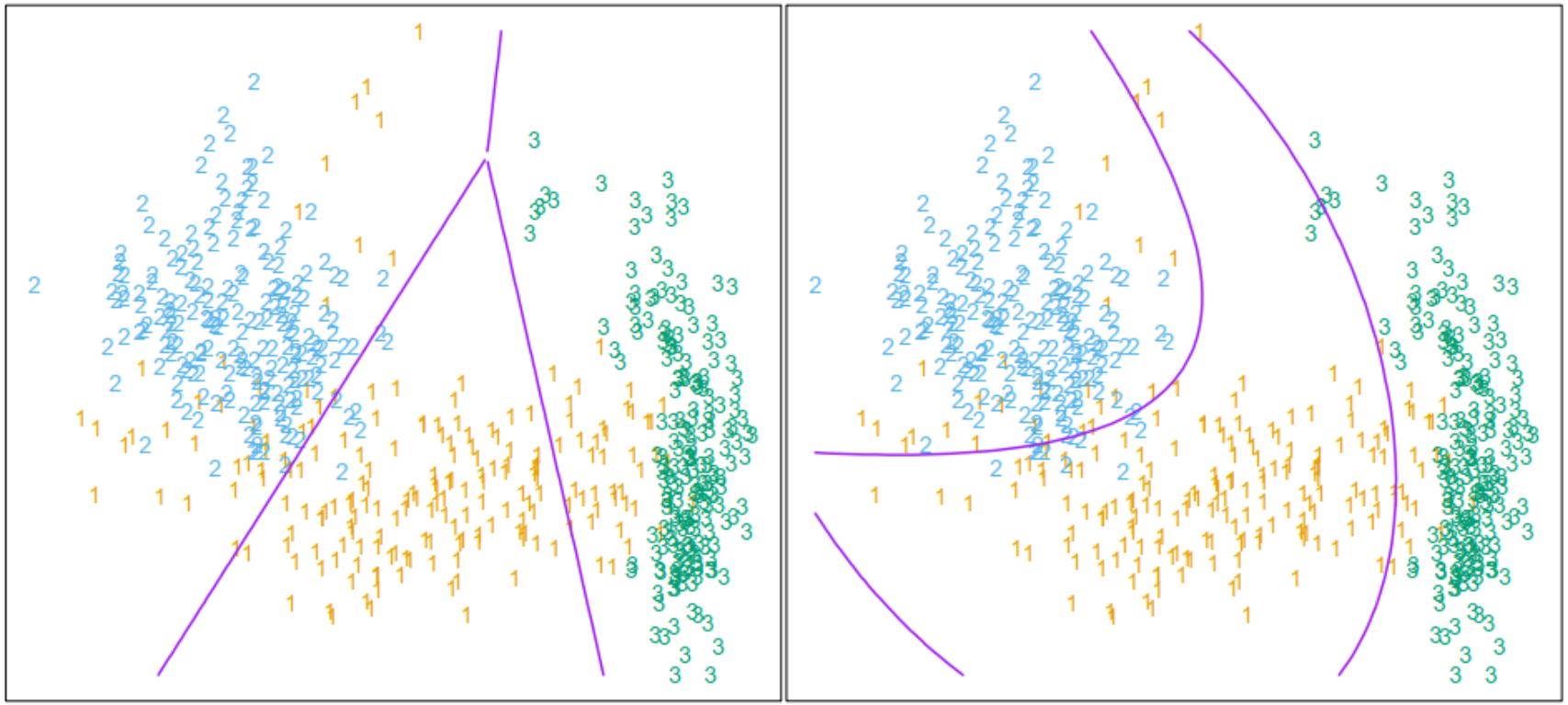
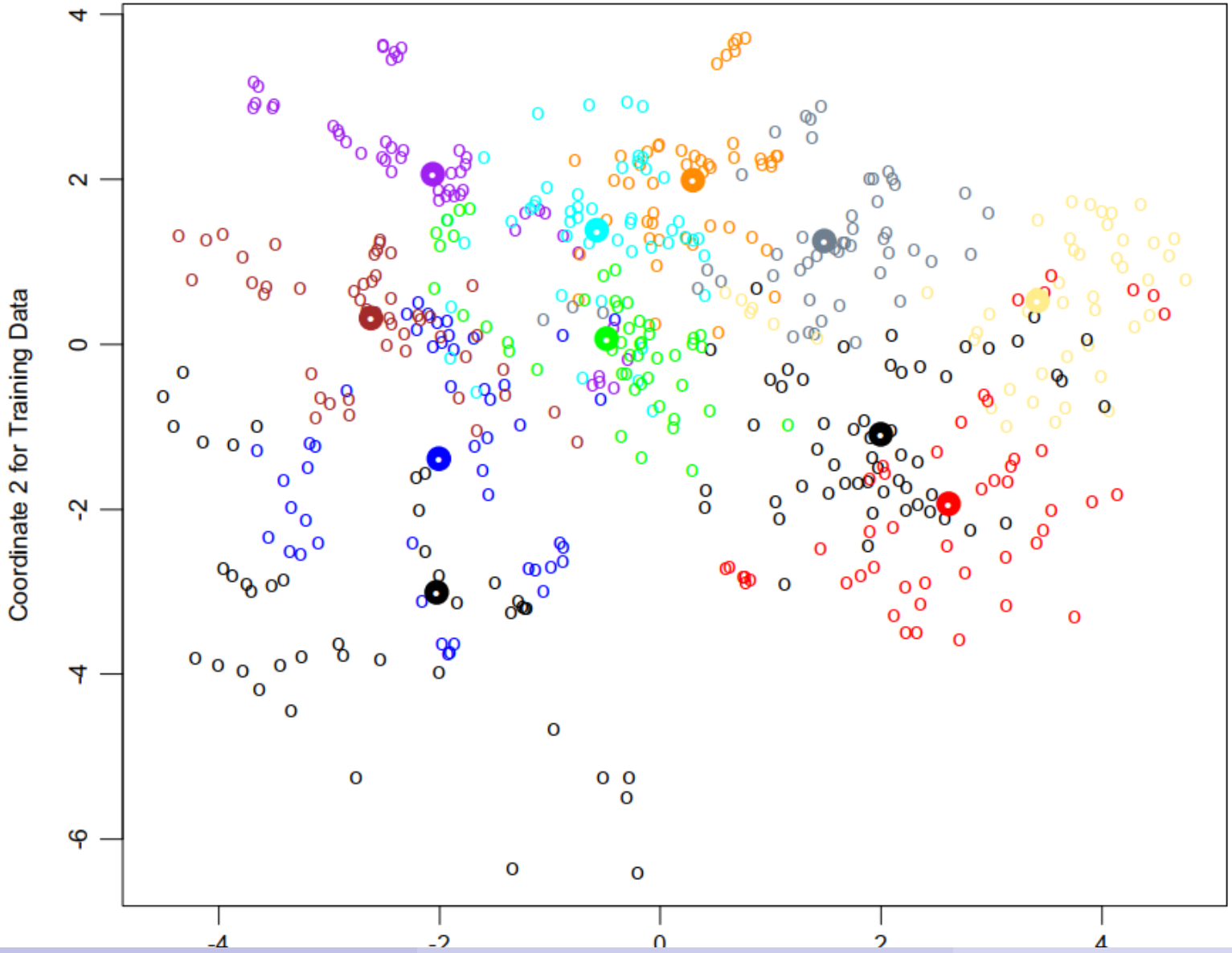


FIGURE 4.1. *The left plot shows some data from three classes, with linear decision boundaries found by linear discriminant analysis. The right plot shows quadratic decision boundaries. These were obtained by finding linear boundaries in the five-dimensional space* $X_1, X_2, X_1X_2, X_1^2, X_2^2$. *Linear inequalities in this space are quadratic inequalities in the original space.*

Another example on vowel data



Logistic Regression

Binary case: $Y \in \{0, 1\}$, where the model is

$$P(Y = 1|X = x) = \frac{1}{1 + e^{-w^T x}}$$

Then the log-likelihood is written as

$$\ell(w) = \sum_{i=1}^{n} \left[y_i w^T x_i - \log(1 + e^{w^T x_i}) \right]$$

This convex optimization problem (unique global minimum) can be solved to find the best values of w.

Intuition and implications:

- Directly models $P(Y|X)$, avoiding assumptions about generating X.
- Logistic function converts $w^T X$ into a probability between 0 and 1.
- The decision boundary is a hyperplane defined by $w^T x = 0$.

More details

$$\beta^{\text{new}} = \beta^{\text{old}} - \left(\frac{\partial^2 \ell(\beta)}{\partial\beta\partial\beta^T}\right)^{-1} \frac{\partial \ell(\beta)}{\partial\beta},$$

Let W be a $N \times N$ diagonal matrix of weights with the ith diagonal element $p(x_i; \beta_{old})(1 - p(x_i; \beta_{old}))$.

$$\begin{aligned}
\beta^{\text{new}} &= \beta^{\text{old}} + (\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}\mathbf{X}^T(\mathbf{y} - \mathbf{p}) \\
&= (\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{W}\left(\mathbf{X}\beta^{\text{old}} + \mathbf{W}^{-1}(\mathbf{y} - \mathbf{p})\right) \\
&= (\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{W}\mathbf{z}.
\end{aligned}$$

Logistic regression models are used mostly as a data analysis and inference tool, where the goal is to understand the role of the input variables in explaining the outcome.

Typically, many models are fit in a search for a parsimonious model involving a subset of the variables, possibly with some interactions terms.

Interpretation of Logistic Regression

Rewrite the model in odds form:

$$\frac{P(Y=1|X=x)}{P(Y=0|X=x)} = e^{w^T x}$$

Taking logs:

$$\log \frac{P(Y=1|X=x)}{P(Y=0|X=x)} = w^T x$$

Intuition and implications:

- Models **log-odds** as a linear fn of x, giving a clear statistical interpretation to the coefficients.
- w represents the change in log-odds for a unit change in the feature.
- Interpretability: +ve weight implies higher prob of class 1 and so on.
- Disadvantage: complex nonlinear relationships cannot be captured without feature engineering or kernels.

Regularized Logistic Regression

Add penalty:

$$\min_w -\ell(w) + \lambda\|w\|^2 \quad \text{or} \quad \min_w -\ell(w) + \lambda\|w\|_1$$

Notes:

- To control model complexity: penalize large coefficients, prevent overfitting in high-dimensional or noisy settings.
- L2 regularization shrinks coefficients smoothly and improves numerical stability, whereas L1 regularization promotes sparsity
- This connects directly to earlier regression frameworks: logistic regression with regularization is the classification analogue of ridge and lasso.
- The tradeoff parameter λ balances data fit and model complexity, reflecting the bias–variance tradeoff.

An example on vowel data (contd)

TABLE 4.1. *Training and test error rates using a variety of linear techniques on the vowel data. There are eleven classes in ten dimensions, of which three account for 90% of the variance (via a principal components analysis). We see that linear regression is hurt by masking, increasing the test and training error by over 10%.*

Technique	Error Rates	
	Training	Test
Linear regression	0.48	0.67
Linear discriminant analysis	0.32	0.56
Quadratic discriminant analysis	0.01	0.53
Logistic regression	0.22	0.51

Another example

TABLE 4.2. *Results from a logistic regression fit to the South African heart disease data.*

	Coefficient	Std. Error	Z Score
(Intercept)	−4.130	0.964	−4.285
sbp	0.006	0.006	1.023
tobacco	0.080	0.026	3.034
ldl	0.185	0.057	3.219
famhist	0.939	0.225	4.178
obesity	-0.035	0.029	−1.187
alcohol	0.001	0.004	0.136
age	0.043	0.010	4.184

Logistic vs LDA: Same Form, Different Assumptions

LDA log-posterior odds:

$$\log \frac{\Pr(G = k|X = x)}{\Pr(G = K|X = x)} = \alpha_{k0} + \alpha_k^T x$$

Logistic regression:

$$\log \frac{\Pr(G = k|X = x)}{\Pr(G = K|X = x)} = \beta_{k0} + \beta_k^T x$$

- The difference is not in form, but in **how parameters are estimated**.
- Logistic regression maximizes **conditional likelihood** $P(G|X)$ and ignores the distribution of X. Thus, more flexible, as it avoids strong assumptions about the input distribution.
- LDA maximizes **joint likelihood** $P(X, G)$, including the marginal $P(X)$. LDA imposes structure on the data, thereby improving estimation.

Efficiency vs Robustness: Logistic vs LDA

- By modeling $P(X)$, LDA uses more information and can estimate parameters more efficiently (lower variance).
- If the Gaussian assumption is correct, LDA can achieve significantly better performance with less data.
- Logistic regression discards information about $P(X)$, making it less efficient but more robust.
- LDA is sensitive to outliers and model misspecification because it relies on strong distributional assumptions.
- Logistic regression is more stable in practice, especially with non-Gaussian or mixed-type features.
- LDA remains well-defined even in cases where logistic regression breaks down (e.g., perfect separation).
- Use LDA if Gaussian structure is plausible and data is limited. Use logistic regression as a default when assumptions are unclear.

Separating Hyperplanes

General linear classifier:

$$f(x) = x^T \beta + \beta_0$$

Decision rule:

$$\text{sign}(f(x))$$

- A hyperplane $w^T x + b = 0$ divides the space into two half-spaces, each corresponding to a class.
- The vector w determines the orientation of the boundary, while b shifts it.
- Classification reduces to checking on which side of the hyperplane a point lies.
- If the data is linearly separable, we can find (w, b) such that

$$y_i(x_i^T \beta + \beta_0) > 0 \quad \forall i$$

- However, many such separating hyperplanes exist; the question is which one generalizes best.

From Separating Hyperplanes to Margin Maximization

Given data (x_i, y_i) with $y_i \in \{-1, 1\}$, consider a hyperplane:

$$f(x) = x^T \beta + \beta_0$$

Classification rule:

$$G(x) = \text{sign}(f(x))$$

- When classes are linearly separable, many hyperplanes achieve zero training error.
- The question is not separability, but *which separating hyperplane is optimal*.
- The signed value $f(x)$ represents the distance from the hyperplane (after normalization).
- This leads naturally to the idea of choosing the hyperplane that is farthest from the data.

Margin and Optimal Separating Hyperplane

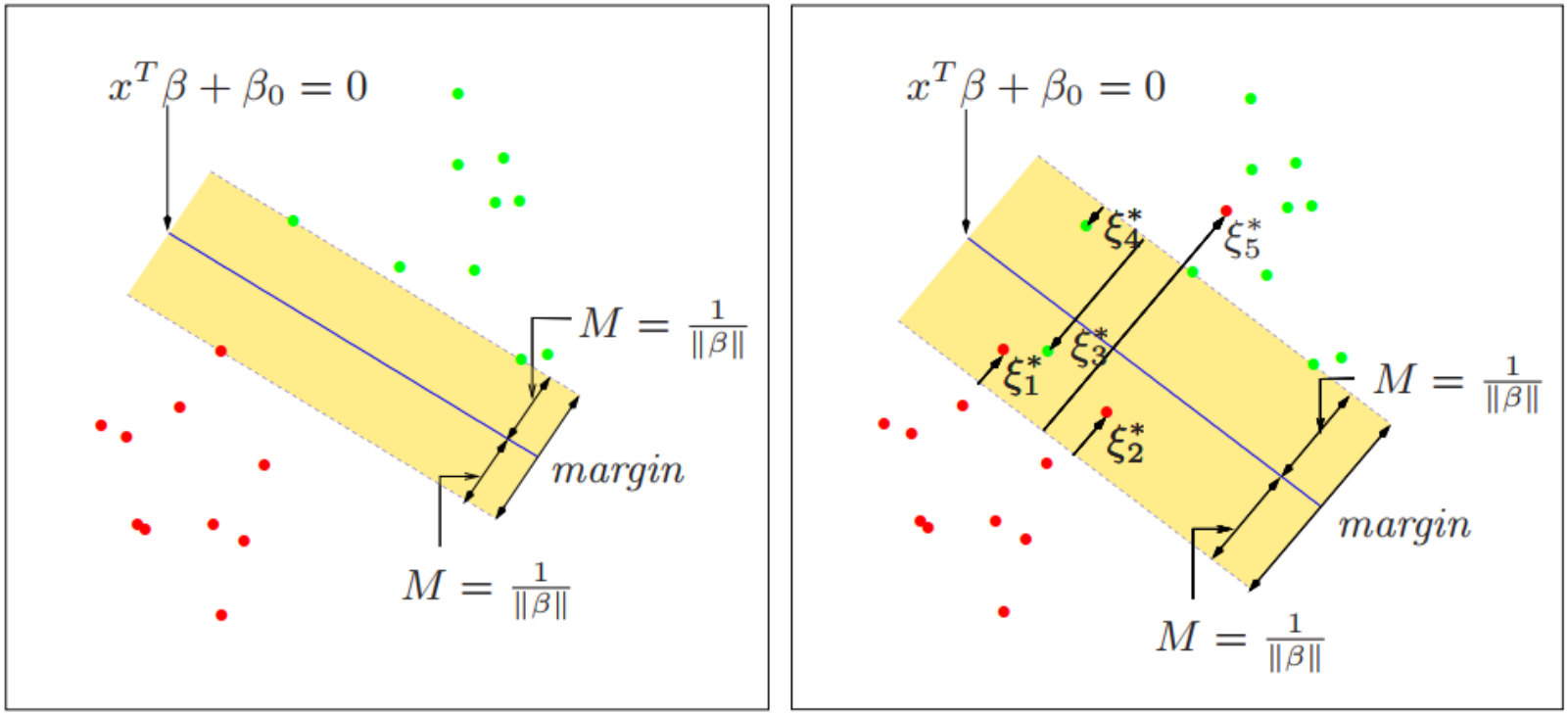


FIGURE 12.1. *Support vector classifiers. The left panel shows the separable case. The decision boundary is the solid line, while broken lines bound the shaded maximal margin of width* $2M = 2/\|\beta\|$. *The right panel shows the nonseparable (overlap) case. The points labeled* ξ_j^* *are on the wrong side of their margin by an amount* $\xi_j^* = M\xi_j$; *points on the correct side have* $\xi_j^* = 0$. *The margin is maximized subject to a total budget* $\sum \xi_i \leq$ *constant. Hence* $\sum \xi_j^*$ *is the total distance of points on the wrong side of their margin.*

Margin and Optimal Separating Hyperplane

Define the margin M:

$$y_i(x_i^T\beta + \beta_0) \geq M, \quad \|\beta\| = 1$$

- The margin is the minimum distance of any training point to the decision boundary.
- The band shown in the figure has width $2M$, enclosing the largest empty region between classes.
- Maximizing M leads to the most stable separation under perturbations of the data.

Optimization problem:

$$\max_{\beta,\beta_0} M \quad \text{s.t. } y_i(x_i^T\beta + \beta_0) \geq M$$

Equivalent formulation:

$$\min_{\beta,\beta_0} \|\beta\| \quad \text{s.t. } y_i(x_i^T\beta + \beta_0) \geq 1$$

- The scale is fixed by the constraint; then $M = 1/\|\beta\|$.

When Classes Overlap

In practice, perfect separation is rare.

- Some points will lie inside the margin or even on the wrong side.
- Enforcing strict separation leads to overfitting or infeasibility.

Introduce slack variables $\xi_i \geq 0$:

$$y_i(x_i^T \beta + \beta_0) \geq M(1 - \xi_i)$$

- $\xi_i = 0$: correctly classified and outside margin
- $0 < \xi_i < 1$: inside margin but correctly classified
- $\xi_i > 1$: misclassified
- The quantities ξ_i correspond to the violations shown in the figure.
- They measure how far a point lies on the wrong side of its margin.

Controlling Violations: Budget on Slack

We impose a global constraint:

$$\sum_{i=1}^{N} \xi_i \leq C$$

- This bounds the total amount of margin violation.
- It limits both the number and severity of misclassified points.
- Misclassifications correspond to $\xi_i > 1$, so the sum controls total error.
- The model no longer insists on perfect separation.
- Instead, it balances:
  - large margin
  - small total violation

This leads to a controlled relaxation of the hard-margin problem.

Soft-Margin Support Vector Classifier

Final optimization problem:

$$\min_{\beta,\beta_0} \|\beta\| \quad \text{s.t. } y_i(x_i^T\beta + \beta_0) \geq 1 - \xi_i$$

$$\xi_i \geq 0, \quad \sum \xi_i \leq C$$

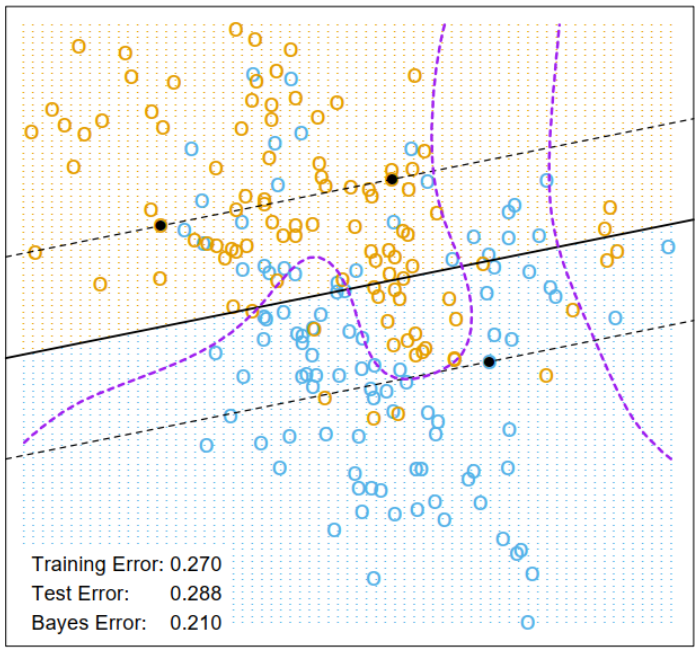
- This is a convex optimization problem (quadratic objective, linear constraints).
- The parameter C controls the tradeoff:
  - small C: allow more violations, wider margin
  - large C: penalize violations, narrower margin
- The solution selects a hyperplane that balances fit and robustness.

Role of Points Near the Boundary

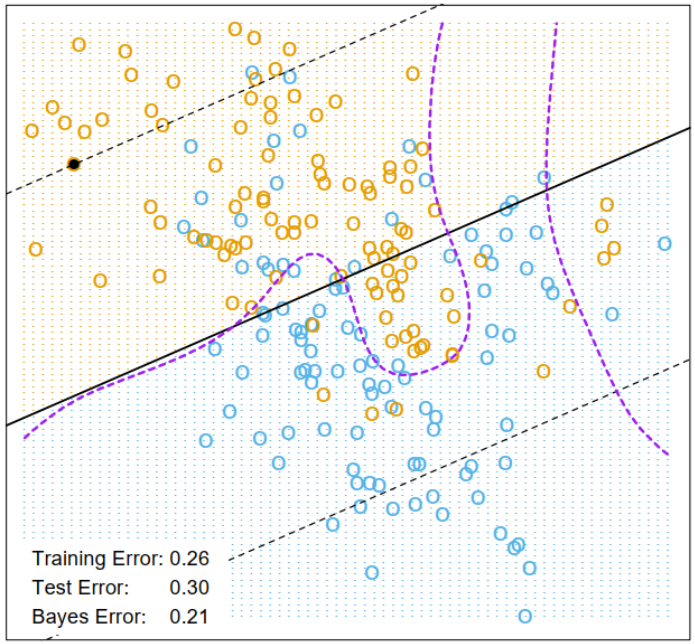
- Only points close to the margin influence the solution.
- Points far inside their class region do not affect the boundary.
- These critical points are called **support vectors**.
- They determine the position and orientation of the decision boundary.
- This is fundamentally different from methods like LDA:
  - LDA depends on global statistics (means, covariance)
  - SVM depends only on boundary points
- The classifier focuses on difficult or ambiguous observations.
- This leads to robustness and good generalization in high dimensions.

Exact computations can be find in the textbook.

An example



$C = 10000$



$C = 0.01$

FIGURE 12.2. *The linear support vector boundary for the mixture data example with two overlapping classes, for two different values of C. The broken lines indicate the margins, where $f(x) = \pm 1$. The support points ($\alpha_i > 0$) are all the points on the wrong side of their margin. The black solid dots are those support points falling exactly on the margin ($\xi_i = 0$, $\alpha_i > 0$). In the upper panel 62% of the observations are support points, while in the lower panel 85% are. The broken purple curve in the background is the Bayes decision boundary.*

From Linear to Nonlinear Decision Boundaries

The support vector classifier constructs a linear boundary:

$$f(x) = x^T \beta + \beta_0$$

- This is restrictive when the true class boundary is nonlinear.
- A natural idea: transform inputs using features

$$h(x) = (h_1(x), \ldots, h_M(x))$$

  and apply the same linear method:

$$f(x) = h(x)^T \beta + \beta_0$$

- A linear boundary in feature space corresponds to a nonlinear boundary in input space.
- The figure illustrates how nonlinear boundaries arise from linear separation after transformation.
- Increasing feature dimension improves flexibility but raises computational and overfitting concerns.

An example - non-linear boundary

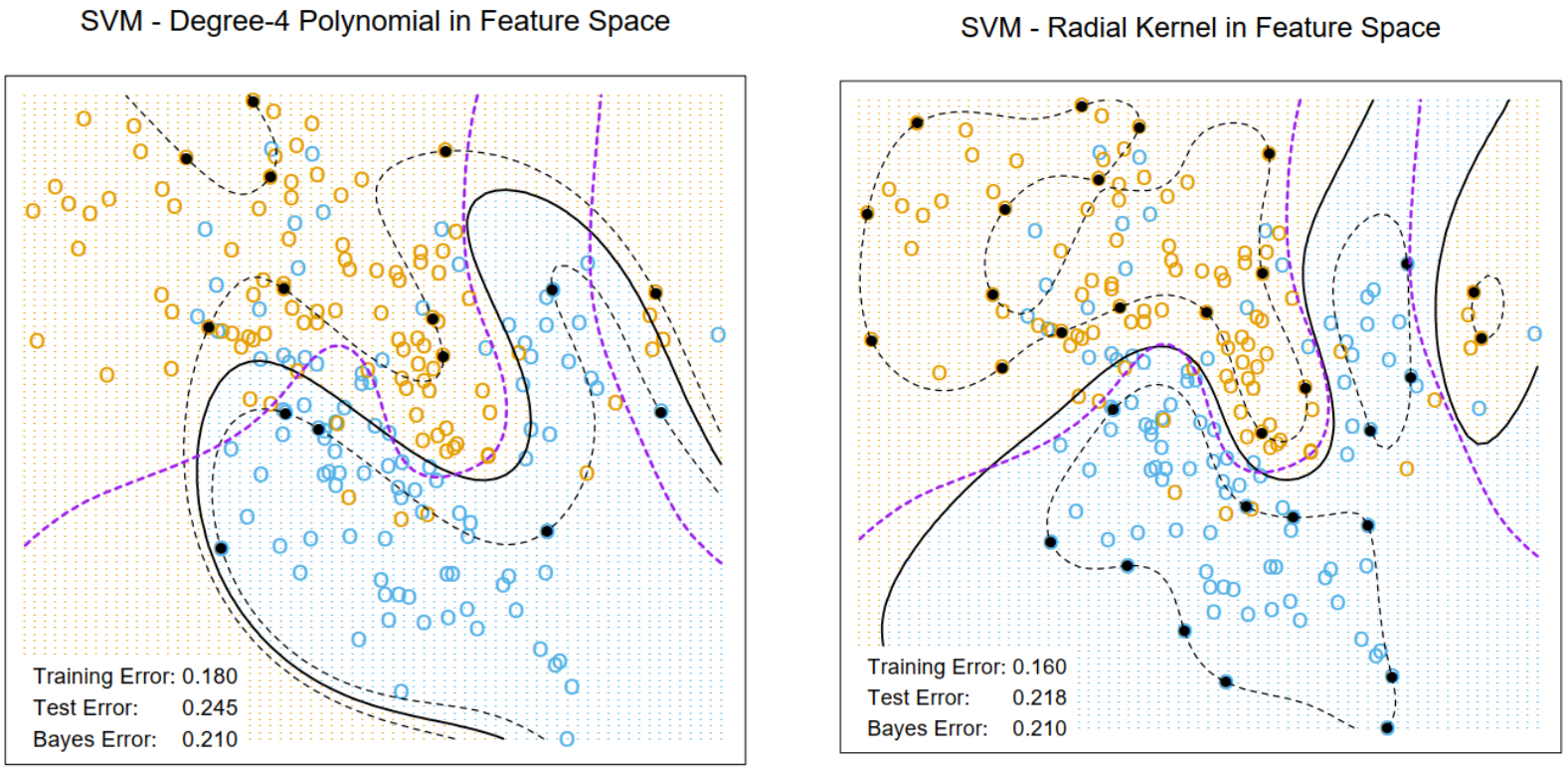


FIGURE 12.3. *Two nonlinear SVMs for the mixture data. The upper plot uses a 4th degree polynomial kernel, the lower a radial basis kernel (with $\gamma = 1$). In each case C was tuned to approximately achieve the best test error performance, and $C = 1$ worked well in both cases. The radial basis kernel performs the best (close to Bayes optimal), as might be expected given the data arise from mixtures of Gaussians. The broken purple curve in the background is the Bayes decision boundary.*

Dual Representation and Inner Products

The SVM solution can be written as:

$$f(x) = \sum_{i=1}^{N} \alpha_i y_i \langle h(x), h(x_i) \rangle + \beta_0$$

- The solution depends on data only through inner products:

$$\langle h(x), h(x_i) \rangle$$

- The explicit form of $h(x)$ is not required.
- Computation is governed entirely by pairwise similarities between points.
- This is a key structural result: the classifier lies in the span of training data.
- Only points with $\alpha_i \neq 0$ (support vectors) contribute.

Kernel Trick and Function Representation

Define the kernel:

$$K(x, x') = \langle h(x), h(x') \rangle$$

Then the classifier becomes:

$$f(x) = \sum_{i=1}^{N} \alpha_i y_i K(x, x_i) + \beta_0$$

- No need to compute $h(x)$ explicitly; all computations use $K(x, x')$.
- Allows implicit mapping into high (even infinite) dimensional spaces.

Examples:

$$K(x, x') = (1 + x^T x')^d, \quad \exp(-\gamma \|x - x'\|^2)$$

- The kernel determines the geometry of the feature space.
- The parameter C still controls smoothness:
  - large C: complex, wiggly boundary (risk of overfitting)
  - small C: smoother boundary

Unifying View

$$\min_f \sum_{i=1}^{n} L(y_i, f(x_i)) + \lambda \|f\|^2$$

- Most supervised learning methods fit into this framework.
- The loss function L encodes how errors are penalized.
- The regularization term controls model complexity.
- The function space (linear or kernel) determines expressiveness.

Examples:

- Squared loss $\rightarrow$ regression
- Logistic loss $\rightarrow$ probabilistic classification
- Hinge loss $\rightarrow$ margin-based classification

Loss Table

TABLE 12.1. *The population minimizers for the different loss functions in Figure 12.4. Logistic regression uses the binomial log-likelihood or deviance. Linear discriminant analysis (Exercise 4.2) uses squared-error loss. The SVM hinge loss estimates the mode of the posterior class probabilities, whereas the others estimate a linear transformation of these probabilities.*

Loss Function	$L[y, f(x)]$	Minimizing Function
Binomial Deviance	$\log[1 + e^{-yf(x)}]$	$f(x) = \log \frac{\Pr(Y = +1\|x)}{\Pr(Y = \text{-}1\|x)}$
SVM Hinge Loss	$[1 - yf(x)]_+$	$f(x) = \text{sign}[\Pr(Y = +1\|x) - \frac{1}{2}]$
Squared Error	$[y - f(x)]^2 = [1 - yf(x)]^2$	$f(x) = 2\Pr(Y = +1\|x) - 1$
"Huberised" Square Hinge Loss	$-4yf(x)$, $yf(x) < \text{-}1$ $[1 - yf(x)]_+^2$ otherwise	$f(x) = 2\Pr(Y = +1\|x) - 1$

Support Vector Regression: Formulation

We now consider regression with a linear model:

$$f(x) = x^T \beta + \beta_0$$

Estimate parameters by minimizing:

$$\sum_{i=1}^{N} V_\epsilon(y_i - f(x_i)) + \frac{\lambda}{2}\|\beta\|^2$$

where the ϵ-insensitive loss is:

$$V_\epsilon(r) = \begin{cases} 0 & |r| < \epsilon \\ |r| - \epsilon & \text{otherwise} \end{cases}$$

- Errors smaller than ϵ are ignored completely.
- Only deviations larger than ϵ contribute to the loss.
- This creates a **tube** of width 2ϵ around the regression function.
- Analogy with classification: points far from the boundary are ignored.
- Here, points with small residuals are ignored.

Properties of the ϵ-Insensitive Loss

- The loss is flat near zero and linear beyond ϵ.
- This differs from squared loss, which penalizes all deviations.
- Small residuals do not affect the fit $\Rightarrow$ robustness to noise.
- Large residuals are penalized linearly $\Rightarrow$ reduced sensitivity to outliers.
- Compared to Huber loss:
  - Huber transitions from quadratic to linear.
  - SVR loss is *flat* near zero and then linear.
- The parameter ϵ controls the width of the insensitive region.
- Larger $\epsilon \Rightarrow$ simpler (flatter) function.

Dual Representation and Kernel Extension

The solution has the form:

$$\hat{f}(x) = \sum_{i=1}^{N} (\hat{\alpha}_i^* - \hat{\alpha}_i)\langle x, x_i \rangle + \beta_0$$

- Only a subset of points have nonzero coefficients.
- These points are the **support vectors**.
- As in classification, the solution depends only on inner products $\langle x, x_i \rangle$
- This allows kernelization:

$$f(x) = \sum_{i=1}^{N} (\hat{\alpha}_i^* - \hat{\alpha}_i) K(x, x_i) + \beta_0$$

- The parameter λ controls smoothness (regularization).
- The pair (ϵ, λ) controls tolerance to small errors and complexity of the function.

Extensions of LDA

Flexible Discriminant Analysis (FDA)

- Applies LDA after transforming inputs via basis functions $h(x)$.
- Produces nonlinear decision boundaries via linearity in feature space.

Penalized Discriminant Analysis (PDA)

- Introduces regularization into discriminant analysis.
- Stabilizes estimation in high dimensions or with correlated features.

Mixture Discriminant Analysis (MDA)

- Models each class as a mixture of distns. rather than a single distn.
- Captures multimodality and yields more flexible decision boundaries.

Summary:

- LDA $\rightarrow$ linear + Gaussian
- FDA $\rightarrow$ nonlinear features
- PDA $\rightarrow$ regularized estimation
- MDA $\rightarrow$ richer class distributions

Summary

- **Two approaches:**
  - LDA models $P(X|Y)$ (generative), logistic regression models $P(Y|X)$ (discriminative)
  - Same linear form, different assumptions and estimation
- **Linear classification:**
  - Decision boundaries are hyperplanes $w^T x + b = 0$
  - Regularization controls complexity (ridge / lasso analogue)
- **Margin viewpoint (SVM):**
  - Choose separating hyperplane with maximum margin
  - Soft margins allow violations, controlled by C
- **Kernel methods:**
  - Replace $\langle x, x' \rangle$ with $K(x, x')$
  - Linear methods $\Rightarrow$ nonlinear boundaries in input space
- **Unifying idea:**

$$\text{Loss} + \text{Regularization} + \text{Feature space}$$